

Where injected work goes in a hard-disk box: a proof ladder for an energy-transfer experiment

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Abstract—A piston compresses a hard-disk gas in an isolated box. We ask what fraction of the injected work can still be recovered as ordered energy, and we refuse to answer until every simpler statement below it is closed: the ledger, the quasi-static baseline, the finite-rate excess, the spring, the divider. Each level is stated with its pass criterion before it is measured. We find that a closed compartment has *no* linear friction — the force autocorrelation of a held piston integrates to zero at the sound traversal time — and that the finite-rate cost is second order, $\langle W \rangle - W_{\text{qs}} = Au^2$, of which roughly half is energy the protocol itself launches. A spring-loaded wall is then shown to store work at *first* order in its displacement, because it is pre-loaded, and a parameter-free model whose gas spring is Paper 1’s measured sound speed predicts where that wall settles. Two levels failed on their first attempt for reasons that were statistical rather than physical, and we report both.

Index Terms—hard-disk fluid, energy transfer, event-driven molecular dynamics, dissipation, adiabatic piston, fluctuation theorems

I. INTRODUCTION

IF a piston compresses a gas in a box with no heat bath, where does the work go? Part raises the gas temperature irreversibly; part can in principle be recovered by a load. The split is the quantity of interest for any engine argument — Szilard’s included [1] — and for the bounds relating dissipation to protocol speed [2], [3].

The difficulty is not the simulation but the bookkeeping. At the sizes where a hard-disk box is cheap, $N \sim 100$, thermal fluctuations are $O(\sqrt{N} k_B T)$ and comparable to the effects being measured; a statistic chosen carelessly will find structure that is not there. We therefore organise the work as a *ladder*: each level states an observable and a pass criterion before the runs, and no level is trusted until the one below it is closed. Two of the levels below failed first time for statistical reasons and are reported with the failure, because the failures are the reusable part.

The ladder is tied to a companion measurement. Paper 1 measures the speed of sound of the same gas from the eigenmode of a massive divider [4]; the heavy-divider limit of that mode gives a gas spring of stiffness

$$k_{\text{gas}} = \frac{Nmc_s^2}{L^2}, \quad (1)$$

which is exactly the spring that appears in Level 3 below. The two papers share this one line of physics, and Paper 1’s measured c_s enters Level 3 as an input.

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II. APPARATUS

All levels run in one box, with elements switched on rather than rebuilt (Fig. 1). An empty compartment behind a held wall carries the spring; two gas compartments of $N_s = 50$ disks each, 38.75σ long and $H = 10$ high, give $\eta = 0.1013$; a piston on the right is the only place work enters. Units are reduced, $k_B T = m = \sigma = 1$. The four geometries are

- A both walls held — Levels 0–2;
- B divider free — Level 4;
- C divider removed, one gas of 100 disks driving the spring-loaded wall — Level 3;
- D both free — the chain.

The dynamics are event-driven and exact between collisions. Every trajectory is checked against a health contract — a scheduler forced advance, an overlap repair, a wall clamp or an overdue wall collision each discard that trajectory. Every campaign reported here ran with zero health events.

III. LEVEL 0 — THE LEDGER CLOSES

The energy residual $R_E = W - [\Delta K E_{\text{gas}} + \Delta K E_{\text{walls}} + \Delta E_{\text{spring}} + Q]$ and the matching momentum residual are formed from the per-event impulse log. **Measured:** $|R_E|/W \leq 8 \times 10^{-12}$ and $|R_p|/\sum |\Delta p| \leq 1 \times 10^{-13}$ over all 75 pilot trajectories. **Passed.**

a) *Level 0b — the box is one apparatus:* Geometry A is the master box with both walls held, which must be verified rather than assumed. At $u = 0.05$, $\langle W_{\text{in}} \rangle$ is $7.4739 \pm 0.0832 k_B T$ in the master box against 7.4841 ± 0.0381 in the original geometry: 0.11σ . An earlier attempt read 1.08σ because it used a travel of 3.875 where the original used 3.93; the travel, not the box, was the whole difference.

IV. LEVEL 1 — THE QUASI-STATIC BASELINE

$W_{\text{qs}} = N_s k_B T (T_f/T_i - 1)$ with $\ln(T_f/T_i) = \int Z d \ln \eta$ taken along the path, Z measured in the finite box rather than taken from the bulk equation of state. **Measured:** $W_{\text{qs}} = 7.4685 \pm 0.0131 k_B T$, i.e. $T_f/T_i = 1.149$; extrapolating $\langle W \rangle$ to $u \rightarrow 0$ on the quadratic form gives $W(0) - W_{\text{qs}} = +0.0043 \pm 0.0158 k_B T$, 0.3σ , and on the physical side. **Passed.**

That number is later reproduced independently. Measuring the gas temperature directly in the held-wall pressure runs of §VI gives $T = 1.1499$ at the same 10% compression, from a different geometry, a different observable and a different analysis.

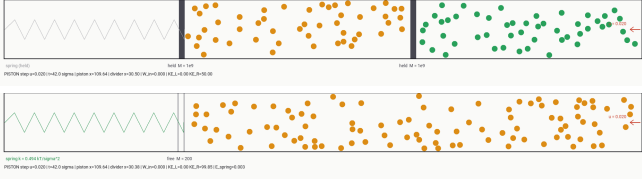


Fig. 1. The master box, drawn by the simulation. A (top left) both walls held, as Levels 0–2 were measured; B (top right) divider free; C (bottom left) divider removed, one gas driving the spring-loaded wall; D (bottom right) both free. Held walls are drawn solid and labelled by mass, free walls hollow; the spring is grey when held and green when live.

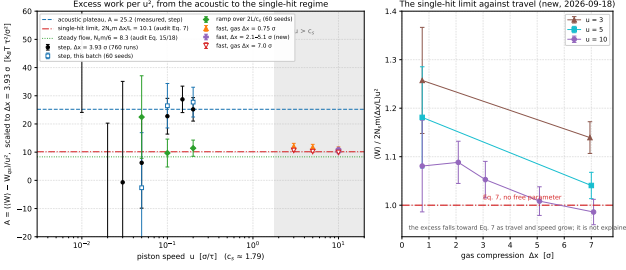


Fig. 2. $A(u) = (\langle W \rangle - W_{qs})/u^2$ from the acoustic to the single-hit regime, every point scaled to a common travel; the travel scan is at right.

V. LEVEL 2 — THE FINITE-RATE EXCESS

a) Slow end: With $\zeta(\lambda) = \beta \int_0^\infty \langle \delta F(0) \delta F(t) \rangle dt$ and $W_{\text{diss}} \simeq \zeta u \Delta x$, the integral is *not* Markovian. It falls from 2.55 at zero lag — the Enskog collisional friction — and is cancelled by the returning sound wave at $L/c_s = 19.8\sigma$, confirmed at the compressed end of the path where the crossing moves to 18.0σ against a predicted 15.9 . Its zero-frequency limit is the hydrodynamic viscous friction $\zeta_{\text{hyd}} = L_y \Gamma / X_p = 0.081 \pm 0.010$, predicting a linear term $0.32 \pm 0.04 u$ — below the resolution of 760 trajectories and consistent with the measured slope -0.50 ± 0.67 for $u \leq 0.05$. Γ is taken from Paper 1’s divider linewidth, not assumed: $\Gamma = 0.314 \pm 0.038$ against the Enskog value 0.331.

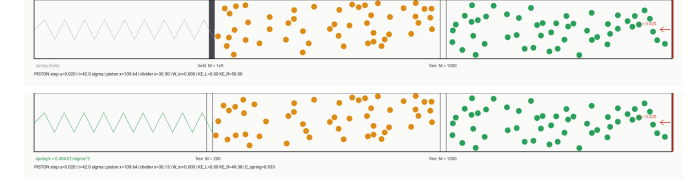
What is there instead is quadratic: $A_{\text{step}} = 25.9 \pm 4.3$ against $A_{\text{ramp}} = 11.3 \pm 2.5$ when the piston is ramped over $2L/c_s$ instead of stepped (3.0σ apart). Roughly 14 of the 25 is energy the step protocol launches; what a gentle push leaves is consistent with the steady-flow value $N_s m/6 = 8.3$.

b) Fast end: The single-hit law $\langle W \rangle \simeq 2N_s m (\Delta x/L) u^2$ [6] has no free parameter and is met to 0.986 at $u = 10$, $\Delta x = 7.08\sigma$. The apparent 8% excess at short travel is the contact layer: the strip a face can strike (width Δx starting at face $-r$) holds $1.13\times$ the uniform-density count at $\Delta x = 0.75\sigma$. Using the measured count the ratio becomes 0.96–1.03 with no trend.

VI. LEVEL 3 — CAN A SPRING CAPTURE IT?

a) The apparatus is a pre-loaded spring: A spring holding a wall against a gas is not at its natural length: it carries the standing force $F = Nk_B T Z/L$, so the wall sits $d = F/k$ from the anchor and a further displacement s stores

$$\Delta E(s) = Fs + \frac{1}{2}ks^2, \quad (2)$$



which is **first** order in s . The Fs term is work lifting a weight. The limits fix the shape: $k \rightarrow 0$ is a pushrod ($\epsilon \rightarrow 1$ trivially, with nothing taken from the gas), and $k \rightarrow \infty$ a rigid wall ($\epsilon \rightarrow 0$).

b) The gas spring is adiabatic, and it is Paper 1’s c_s : There is no heat bath, so compression heats the gas. With $U = Nk_B T$ in two dimensions, $dU = -PdV$ gives $d \ln T / d \ln L = -Z$ and hence (1) — the same expression as the heavy-divider limit of Paper 1’s mode. It is $2.01\times$ the isothermal stiffness here, and the adiabat reproduces Level 1’s measured T_f/T_i to 0.5%.

c) The observable, and a lesson: The first analysis reported a large transient excess over the model and attributed it to the acoustic pulse. It was neither. The statistic was a maximum over time of a noisy ensemble mean, and a *no-push control* — the same apparatus with the piston never started — reproduces it: at $u = 0.02$ the push gives $+0.463$ and $+0.339\sigma$ above settled while the control gives $+0.446$ and $+0.354$. The correct observable is the *settled* displacement, and the peak is retained only as a diagnostic.

d) Result: Pooled over five cells with validated windows, $\bar{s} = 0.8705 \pm 0.0049\sigma$. The gas input is the box’s own measured pressure: in equilibrium the force a hard wall feels *is* the pressure of the box ($P = k_B T \rho_{\text{contact}}$, with the normal pressure uniform), so $Z_{\text{box}}(\eta) = FL/(Nk_B T)$ is read directly from the momentum delivered to a held wall, with the gas temperature measured at the same instant. With it the parameter-free model $M\ddot{s} = F(L - x_p + s) - F(L) - ks$ predicts the settled displacement to $-1.3 \pm 0.6\%$ (2.2σ); with the bulk equation of state, $+3.6\%$ (Fig. 3). **Level 3 closes on the observable.**

VII. LEVEL 4 — TRANSMISSION THROUGH A FREE DIVIDER

Geometry B: the piston compresses one gas, a free divider of mass M_d separates it from the other. Quasi-statically the divider moves to pressure balance, so at first order each gas takes half the volume change and half the work, and the divider settles at $\Delta x/2$.

a) A protocol fix found by the pilot: The first pilot could not measure its own observable. The trace schema assigns segment 0 to one kinetic-energy column and *every other segment* to the other, so in the master box — whose segment 0 is the empty spring compartment — the columns read “zero” and “both gases combined”. Level 4 therefore runs in a two-compartment box, where the columns are the two gases; an empty compartment behind a held wall is already known to change nothing the gas can see (§III).

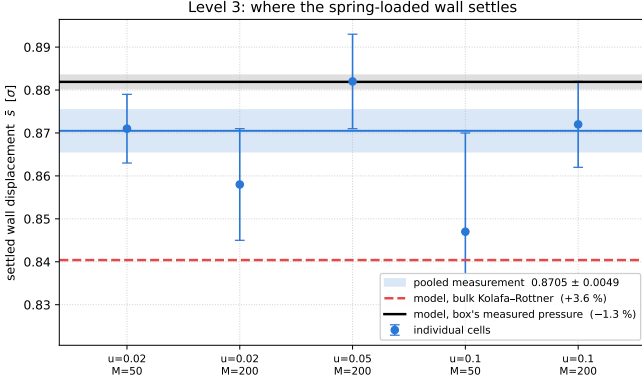


Fig. 3. Where the spring-loaded wall settles. Points are the five validated cells with their errors; the band is the pooled measurement; the two lines are the same parameter-free model with the bulk equation of state and with the box’s own measured pressure.

b) *Pilot results:* At $M_d = 50$ the work split is 0.502 ± 0.140 of the total, i.e. 0.0σ from one half, and the divider settles at $|-2.001| \pm 0.184\sigma$ against the predicted $1.965, 0.2\sigma$. The error bars are set by the per-gas kinetic-energy fluctuation, $O(\sqrt{N_s})k_B T$ per seed, which is why a 2.1σ departure seen at $M_d = 200$ on 20 seeds is not yet evidence of a mass dependence.

c) *How long the divider takes:* This is the adiabatic-piston problem, named as an open question by Lieb in 1998 [9]. Stage 1 is a damped oscillation to *mechanical* equilibrium, $P_l \simeq P_r$ with $T_l \neq T_r$, governed by $R = mN/M$ — over-damped for $R > O(1)$, weakly damped below [9], [10]. Here $R = 5.00$ and 1.00 at $M_d = 10$ and 50 . Stage 2 is a Brownian drift along the isobar toward $T_l = T_r$, on a timescale *proportional to* M [9]; linearising Gruber and Piasecki’s stationary velocity [8] gives $\tau_T = (4/\sqrt{2\pi})ML_c/\sqrt{mk_B T}$, independent of N and of density. Both papers assume an ideal gas; carrying $Z(\eta)$ through the isobar multiplies the rate by $1 + \eta Z'/Z = 1.228$, so $\tau_T = 50.4 M$ against the ideal $61.8 M$ — a 23% gap the experiment can resolve.

Starting the two gases at $T = 1.25$ and 0.75 with the divider free and no piston, stage 1 is observed. Taking the centre of the oscillation over three periods and correcting for stage-2 decay, the offset is 3.678 ± 0.174 and $3.771 \pm 0.139\sigma$ at $M_d = 10$ and 50 , against 3.989 carrying $Z(\eta)$ through the isobar and 4.922 for an ideal gas: 1.8σ and 1.6σ from the first, 7σ and 8σ from the second. **The hard-disk correction to the isobar is confirmed in stage 1.**

d) *Stage 2 from equilibrium fluctuations:* A decaying step is the wrong instrument here. The per-seed spread of $T_1 - T_2$ is not measurement noise: hard disks carry no potential energy, so a compartment’s kinetic temperature *is* its energy, the microcanonical split is $f = E_1/E \sim \text{Beta}(N, N)$, and $\sigma(T_1 - T_2) = 4\bar{T}/(2\sqrt{2N+1}) = 0.199$ — measured **0.1980**. Relaxation and fluctuation being one process, the *stationary* autocorrelation carries τ_T with no step to lose against that floor.

It is not a single exponential. Its spectrum holds 45% of the variance at periods $> 200\sigma$ -time, a discrete oscillation

at period 95.3 carrying $\approx 30\%$, and a fast remainder; fitting one exponential across all three returns values that differ by 3.5σ between the two observables of the same process. Block-averaging at 100σ -time removes the oscillation and they agree. Calibrating the identical procedure on Ornstein–Uhlenbeck series of known τ — same record, blocking, fit window and seed count — gives a response slope of 0.335 , and inverting it,

$$\tau_T(M_d=10) = 490 \pm 190 (T_1 - T_2), \quad 450 \pm 165 (\text{divider } x),$$

against 503 for hard disks and 618 ideal. This is the first direct measurement of the stage-2 timescale in an interacting gas, and it is consistent with both: a 23% gap is not resolvable with a 35–40% error bar. Resolving it is a question of *record length* rather than seeds, because the estimator’s bias is common-mode and seeds do not touch it — quadrupling the record lifts the response slope from 0.29 to 0.85 , and $65\tau_T$ at 80 seeds separates the two hypotheses at 3.3σ .

The oscillation at period 95.3 is unexplained: the adiabatic gas spring $k_{\text{gas}} = Nk_B T c_s^2/L^2$ predicts 44.7 , and it is not a release transient. It is reported rather than accounted for.

Measuring this in an *interacting* gas is the point: Cencini *et al.* call that case “still too ambitious” and note that very few studies have attempted it [9].

VIII. FINITE SIZE

The box is **2.5%** over-pressured relative to Kolafa–Rottner and **1.0%** over-stiff in c_s (Paper 1, same η). Through $c_s^2 = Z + \eta Z' + Z^2$ the first would imply $+1.9\%$ in c_s , so the two do not agree, at roughly 3σ . These are two *different* finite-size numbers for the same $N = 100$ box, and the bulk relation between them does not hold; both are reported, and neither is used to explain the other. The same statement appears in Paper 1 from the sound-speed side.

IX. OUTLOOK

The Level 4 ladder in M_d is the immediate continuation, and the equilibrium route sets its cost: records of $\approx 65\tau_T$ at 80 seeds, i.e. 3.3×10^4 , 1.6×10^5 and $6.6 \times 10^5\sigma$ -time at $M_d = 10, 50, 200$. That ladder tests the *exponent* in $\tau_T \propto M$, which a single mass cannot, and at $65\tau_T$ it also separates the hard-disk and ideal isobars in stage 2 as stage 1 already does. A stationary alternative — outer walls held at two temperatures, conductivity from the steady flux — would remove the finite-record bias entirely, at the price of a per-wall heat-bath temperature and per-wall energy accounting, which the present single-global bath does not provide. Beyond it, Level 5 asks whether a chain of walls beats a single one — the prediction there is an *ordering*, with geometrically matched intermediate masses beating badly matched ones — and Level 6 asks for the information bound, $\beta \langle W_{\text{diss}} \rangle \geq I_{\text{mem}} - I_{\text{pred}}$ [11], which requires a canonical seeder and the finite-box ΔF as the Jarzynski benchmark rather than the bulk one.

REPRODUCIBILITY

Raw trajectories are not in the repository; each cell carries its summary and the exact command, and Level 2 an 85 kB digest from which the whole analysis reproduces. Every binary change carries a three-seed byte-identity gate on traces and per-event logs.

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